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BirdID

A Mobile Application for Bird Species Identification Using Audio Recognition and Machine Learning

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Introduction

Bird song is one of the most reliable indicators of species presence in a given habitat. Identifying bird species by their vocalisations is challenging even for experienced ornithologists, yet it is critical to wildlife conservation and biodiversity monitoring.

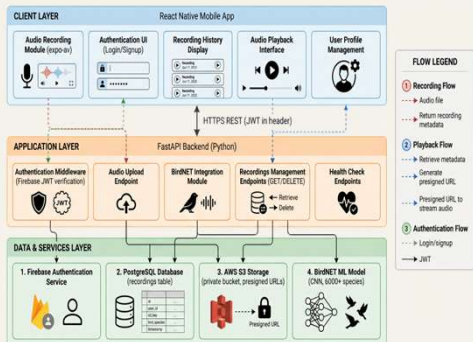
BirdID is a cross-platform mobile application that records bird audio on a smartphone and uses the BirdNET deep learning model to automatically identify the species, returning the common name, scientific name, and a confidence score in real time.



The system integrates five modern cloud services into a cohesive three-tier architecture: Firebase Authentication, AWS S3 object storage, PostgreSQL, Railway cloud hosting, and the Cornell Lab of Ornithology's BirdNET neural network.

The project was developed over six months (Oct 2025 to Apr 2026) following an Agile iterative methodology across 11 development phases, with version control maintained via Git and GitHub throughout.

System Architecture



- Tier 1:** Client Layer: React Native + Expo mobile application with eight screens
- Tier 2:** Application Layer: Python 3.11 FastAPI backend handling JWT authentication, audio format conversion, BirdNET ML inference, AWS S3 upload, and PostgreSQL persistence.
- Tier 3:** Data and Services Layer: Firebase Authentication, PostgreSQL AWS S3, and the BirdNET ML model

Technology Stack

View of the complete **technology stack** including: Frontend, Backend, Data & Storage, and Deployment & DevOps

Frontend (Mobile Application)

- React Native (v0.72+) - Cross-platform mobile framework
- Expo (SDK 49+) - Development platform and tooling
- React Navigation (v6) - Screen navigation and routing
- Firebase SDK (Web) - User authentication
- Expo Linear Gradient - Visual polish and styling

Backend (API Server)

- Python (3.11+) - Programming language
- FastAPI - Modern async web framework
- Unicorn - ASGI server
- SQLAlchemy - ORM for database operations
- Pyrebase - PostgreSQL database adapter
- Boto3 - AWS SDK for S3 operations
- Firebase Admin - Authentication token verification
- BirdNET-Lib - Bird identification ML model
- Pydub - Audio format conversion
- Librosa - Audio processing and analysis

Data & Storage

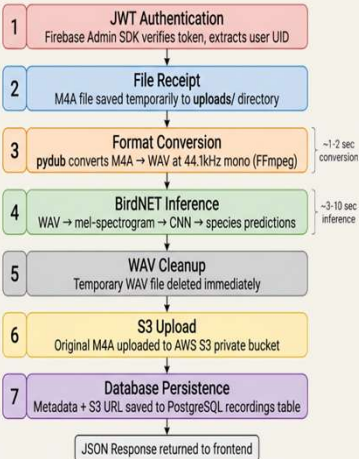
- PostgreSQL (18.0) - Relational database for metadata
- AWS S3 - Cloud object storage for audio files
- Firebase Authentication - User management and JWT tokens

DevOps & Testing

- Git/GitHub - Version control
- GitHub Actions - CI/CD pipeline
- Pytest - Backend testing framework
- Railway - Deployment platform
- Expo Go EAS - Downloadable apk

ML Inference Pipeline

When a user uploads a recording, the backend executes a seven-stage pipeline:



This pipeline allows the user to upload their recording and be redirected to an analysing screen. Analysing takes 3-12 seconds depending on the length of the audio file.

BirdNET Performance on Irish Species

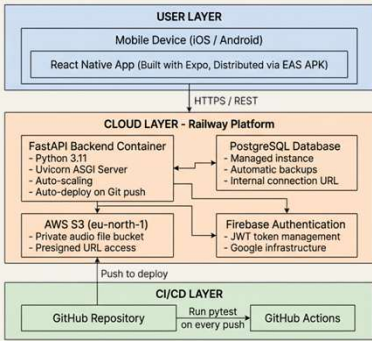


BirdNET is a CNN developed by the **Cornell Lab of Ornithology** and Chemnitz University of Technology, trained on over 6,000 bird species globally. It processes audio spectrograms and returns species predictions with confidence scores. Manual testing on common Irish species produced consistently strong results:

European Robin (<i>Erithacus rubecula</i>)	87 - 93%
Common Blackbird (<i>Turdus merula</i>)	85 - 91%
Song Thrush (<i>Turdus philomelos</i>)	82 - 89%
Blue Tit (<i>Cyanistes caeruleus</i>)	78 - 86%
Great Tit (<i>Parus major</i>)	80 - 88%
Chaffinch (<i>Fringilla coelebs</i>)	76 - 84%
Wren (<i>Troglodytes troglodytes</i>)	74 - 83%
Goldfinch (<i>Carduelis carduelis</i>)	72 - 81%

Overall detection rate: ~80% | Average processing time: ~10 seconds

Deployment Architecture



Scan the QR code to download and use the BirdID app on any Android device.



Project Statistics

~4,300
Lines of Code

150+
Git Commits

15+
Automated Tests

~70%
Test Coverage

8
API Endpoints

6,000+
Bird Species

Testing and CI/CD

API Tests: All 8 endpoints verified; auth rejection (401/403) confirmed on all protected routes.

Integration Tests: BirdNET configuration, database connectivity, Firebase Admin SDK, S3 upload verification.
Slow Tests: Live S3 upload with real bucket.

GitHub Actions CI/CD: Runs on every push to main. Spins up PostgreSQL service container, decodes Firebase credentials from Base64 secret, executes full pytest suite, generates XML coverage report, uploads to Codecov. Build fails automatically on any test failure.



Conclusions

All 12 project objectives were achieved. BirdID is a fully functional, cloud-deployed, ML-powered mobile application integrating 20+ technologies across the complete software stack.

Demonstrating professional software engineering practice: RESTful API design with automatic documentation, multi-layered security architecture, automated CI/CD, cloud-native deployment, and a comprehensive automated test suite with 70% code coverage.

BirdNET achieved 72-93% confidence on common Irish species, validating the core project objective and demonstrating successful ML integration in a student-developed production system.

Future Work



A custom audio similarity algorithm using MFCC feature extraction and cosine similarity for formal comparative evaluation against BirdNET represents the most valuable direction for future development. Further enhancements include the creation of a **GPS map visualisation** of sightings and multiple simultaneous species detection

References & Acknowledgments

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- [3] D. Stowell and M. Plumbley, Automatic large-scale classification of bird sounds, *PeerJ*, vol. 2, 2014.
- [4] R. Fielding, *Architectural Styles and Network-based Software Architectures*, PhD thesis, UC Irvine, 2000.
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- [6] Project Repository: <https://github.com/darragh50/BirdID>